

Global Navigation Satellite System: experiments and outcomes analysis

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ABSTRACT

This report aims to explore GNSS, potential attacks such as spoofing, and inherent vulnerabilities like interference. Measurements were performed under various conditions, followed by an analysis of the data collections. The resulting plots highlight significant outcomes.

1 INTRODUCTION

Knowing our position is a need that also our ancestors had. The instruments to do that have constantly evolved through time, and nowadays we are using **Global Navigation Satellite Systems (GNSS)**. This system is essential for modern positioning systems and, since its creation, it has become increasingly widespread, being embedded into an ever-increasing number of devices and infrastructures. After the deployment of many systems beside Global Positioning System (GPS) and their worldwide adoption and general improvement, the research focus of GNSS has shifted to improving system performance and security [2, 4]. The goal of the paper is to show how GNSS measurements are managed to compute Position, Velocity and Time (PVT) in static, dynamic and uncommon conditions (i.e. strong interference), and to face simulated GNSS spoofing using dedicated software.

2 METHODS

2.1 Device and Software Configuration

In order to perform measurements, a **Samsung Galaxy A33 5G** has been used, running **OneUI 7.0** and **Android 15**. **MATLAB** is adopted to perform data analysis (version R2025b), and **Google GPS Measuring Toolbox** has been equipped with spoofing capabilities modifications (given the configuration, it analyzes only GPS constellation) [3, 5]. All measurements have a duration of 5 minutes and have been performed with **cellular data on** and **battery saving mode** (beside the VOR-DME scenario, where battery saving mode was off, see section 2.3).

2.2 Baseline Scenario

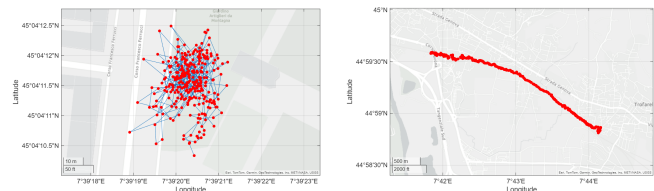
2.2.1 Static Scenario. The log was collected on Thursday 26th March 2026, 11:15:04 UTC +1, at **Comala Association** in Turin (Italy) (45°4'11.892"N 7°39'20.772"E, 252.00 m above sea level), with clear sky conditions (Fig. 1a). The obstacles were mainly made up of a concrete football stand on the East side and sparse trees around the place; it is good to take into account the fact that being in a city means that the full horizon view is obstructed by surrounding buildings.

2.2.2 Dynamic Scenario. The log was collected on Thursday 26th March 2026, 09:35:11 UTC +1, on the **SFM1 26012 train between Trofarello (TO) and Moncalieri (TO)**, with clear sky conditions (Fig. 1b). Obstacles consisted mainly of buildings on the North side, other than the train itself.

2.2.3 Spoofing Conditions. Spoofing has been performed in the static scenario (see section 2.2.1), targeting the **Intesa San Paolo skyscraper summit** (45°4'10.200"N 7°39'46.188"E, 430.00 m above sea level). The spoofed signal starts always after 15 seconds from the start of the log, in order to point out the differences between a first non-spoofed phase and the second spoofed phase of the measurements.

2.3 Interference Scenario

In order to test peculiar conditions and experiment with GNSS interference, two different additional scenarios have been tested. The first is a static scenario in which the receiver is surrounded by strong electromagnetic fields (12 V/m) [7] generated by **Digital Video Broadcasting - Terrestrial (DVB-T)** and **Frequency Modulation (FM) radio antennas** at **Colle della Maddalena** (45°1'52.032"N 7°43'18.300"E, 710.00 m above sea level); the second setting is a static one in which the receiver captures GNSS data close to a **Very-High Frequency Omnidirectional Range (VOR) - Distance Measuring Equipment (DME)** in Poirino (TO) (44°55'31.476"N 7°51'41.976"E, 252.00 m above sea level). Briefly, a VOR-DME is a navigation system for aircraft, emitting and receiving radio signals in the band 960-1215 MHz [6].



(a) Static scenario: Map

(b) Dynamic scenario: Map

Figure 1: Baseline scenario: Maps

3 RESULTS AND DISCUSSION

3.1 Static Scenario

The experimental results for the static scenario confirm several fundamental GNSS characteristics. Although the receiver remained stationary, the observed relative velocities between the device and the satellites were constant but non-null, a result of the combined orbital motion of the space vehicles and the Earth's rotation (Fig. 2b). The signal quality remained quite robust throughout the test; consistent with the theoretical minimum power of approximately

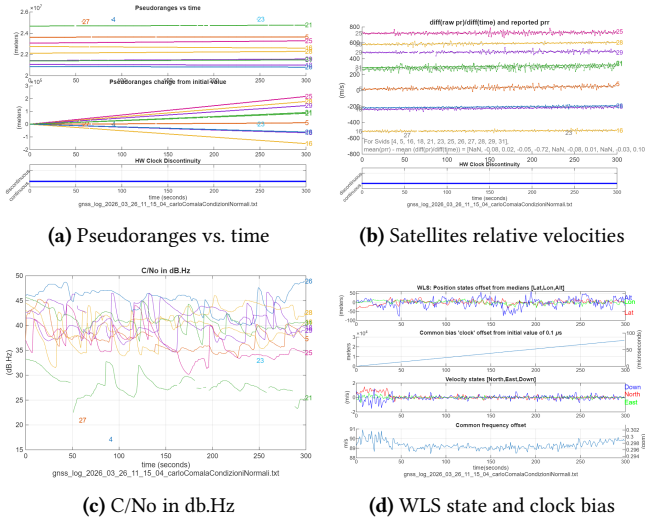


Figure 2: MATLAB outputs for static scenario

-160 dBW for the L1 GPS in-band signals at the Earth's surface [1], the recorded C/N0 values fluctuate between 35 and 50 dB.Hz for the majority of tracked satellites (Fig. 2c). This confirms a clear-sky environment with a standard thermal noise error. Additionally, the receiver clock bias showed a steady, linear increase (Fig. 2d). This accumulation of clock bias error is expected in consumer-grade devices, where the internal clock frequency offset leads to a gradual drift with respect to GNSS precise time.

3.2 Dynamic Scenario

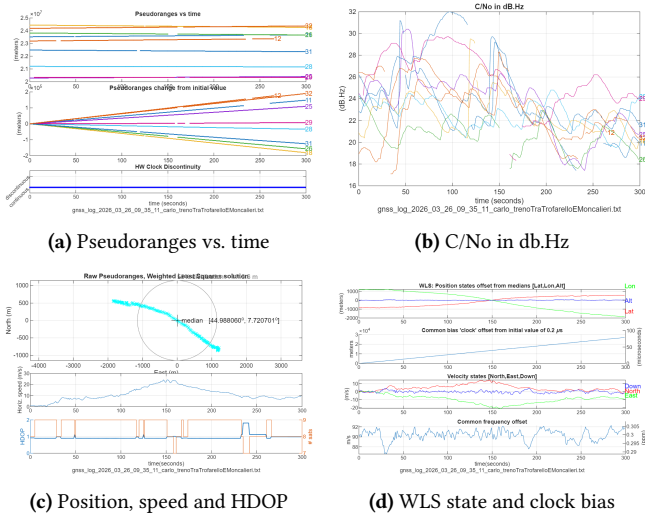


Figure 3: MATLAB outputs for dynamic scenario

In the dynamic scenario, the pseudorange measurements fluctuate in accordance with both the horizontal and vertical displacements of the train and, at the same time, of the receiver (Fig. 3a). A notable decrease in the signal-to-noise ratio (C/N0) was observed, primarily attributed to the electromagnetic shielding provided by the train's

metallic structure, which effectively acted as a Faraday cage and introduced signal interference (Fig. 3b). Despite this attenuation, the receiver successfully tracked the motion, with the trajectory accurately reflecting the simultaneous changes in both position and velocity (Fig. 3c). This demonstrates the system's ability to maintain position and velocity estimation even under non-optimal signal conditions.

3.3 Spoofing

3.3.1 Spoofing Without Delay. At the designated start time (see section 2.2.3), the receiver's position was successfully spoofed to the coordinates of the skyscraper (Fig. 4b). However, the effects of the spoofing attack remained detectable due to significant magnitude spikes in the satellites' relative velocity. These anomalies were triggered by the near-instantaneous shift in the position; specifically, the pseudorange rates exhibited abrupt changes as a result of discontinuous variation in pseudorange values occurring within a single sampling interval (Fig. 4a). These spikes serve as a primary indicator for spoofing detection, highlighting the sudden change in the calculated position.

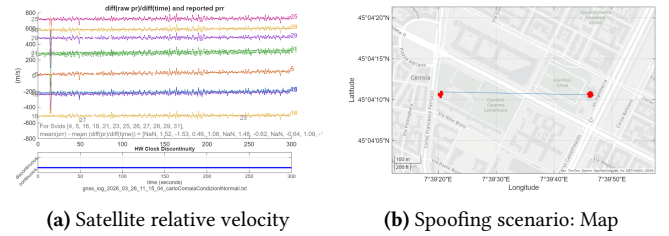


Figure 4: MATLAB outputs for spoofing conditions without delay

3.3.2 Spoofing With Delay. In order to perform spoofing with delay, a small deferral of 1 ms has been injected by the spoofer into the timestamps from all satellites. The attack remains effective, though it introduces a significant discontinuity in the receiver's clock bias. While the reported position remains relatively stable (subject only to minor shifts caused by satellite and Earth rotation during the 1 ms interval), the receiver clock bias shows an instantaneous jump of approximately 1 ms, equivalent to a 300 km pseudorange error. This occurs because the receiver senses its internal time to be ahead of the synchronized atomic clocks of the transmitters. To maintain signal alignment, the device's internal oscillator compensates by adjusting its bias to meet the new timing of the signals. This sharp divergence, clearly visible when comparing the normal clock bias (Fig. 5a) with the delayed case (Fig. 5b).

3.3.3 Spoofing Limitations. In the non-delayed spoofing case, the receiver clock bias maintains its original slope throughout the transition. The device effectively observes an instantaneous position jump without a corresponding shift in its internal clock, creating a clear discrepancy. On the contrary, in the delayed scenario, the receiver treats the satellite atomic clocks as the absolute reference ("truth"). By injecting a common-mode delay, the spoofer forces the receiver to assume "responsibility" for the timing misalignment; the device subsequently adjusts its internal oscillator to match the synthesized signal (Fig. 5). This leads to a silent desynchronization

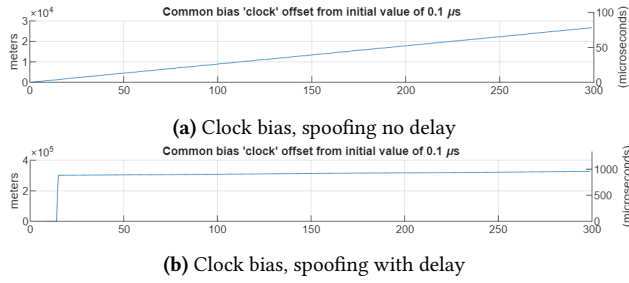


Figure 5: Comparison between clock biases

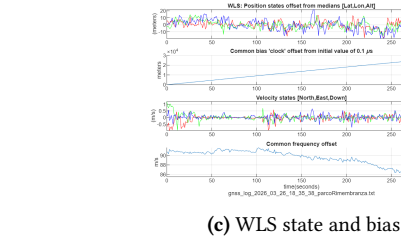
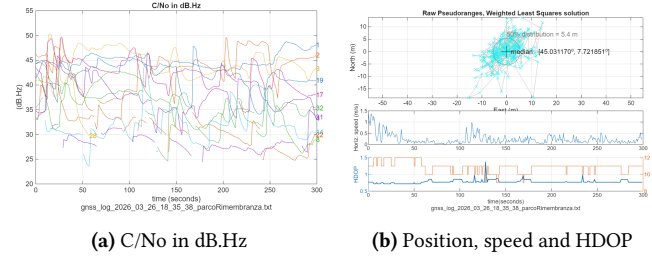


Figure 7: MATLAB outputs for interference scenario, DVB-T/FM

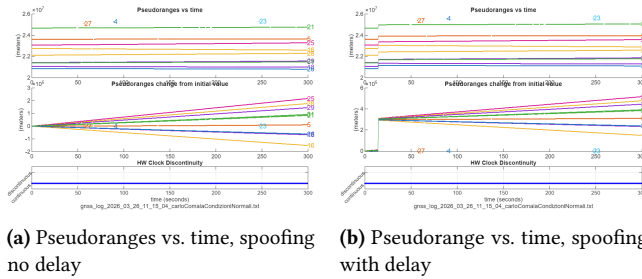


Figure 6: MATLAB outputs for pseudoranges spoofing calculation

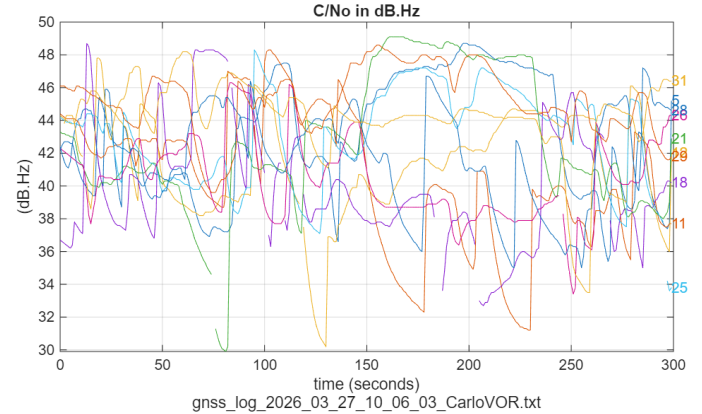


Figure 8: C/No in dB.Hz, VOR-DME interference scenario

3.4 Interference Scenario

3.4.1 DVB-T/FM. During the DVB-T interference test, the receiver maintained a good satellite count due to high elevation and clear-sky visibility (Fig. 7b). However, significant signal degradation was observed across all tracked satellites as a result of the high EM field intensity (Fig. 7a). Beyond simple carrier-to-noise attenuation, a drift in the receiver's frequency offset was recorded (Fig. 7c). This instability is attributed to induced currents within the internal oscillator circuits of the device, caused by the high-power DVB-T and FM transmissions [7]. These currents disrupt the correct functioning of the oscillators (i.e. the clock).

3.4.2 VOR-DME. Regarding VOR-DME, the main concern is about C/No ratio: since DME sends intermitting signals to planes to let them compute relative distance to the VOR, sudden drops are visible at the same time of DME broadcast communications (Fig. 8).

4 CONCLUSION

Globally, it has been possible to get relevant results from selected scenarios and collected data. Particularly, spoofing is a severe threat

to GNSS systems since it not trivial to detect, especially in case of delay injection, leading to clock desynchronization and potential severe conditions. Additionally, GNSS is subjected to RF oscillations, especially near high-power EM sources, leading to general interference, signal degradation and band overlapping. Possible improvements to measurements could include collecting data from Android Developer Mode. This way, the chipset would be forced to track all satellites and frequencies constantly while the GNSS-Logger is active (without it, Android performs duty-cycling to save battery), allowing more accurate and verbose logging.

4 REFERENCES

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5 APPENDIX

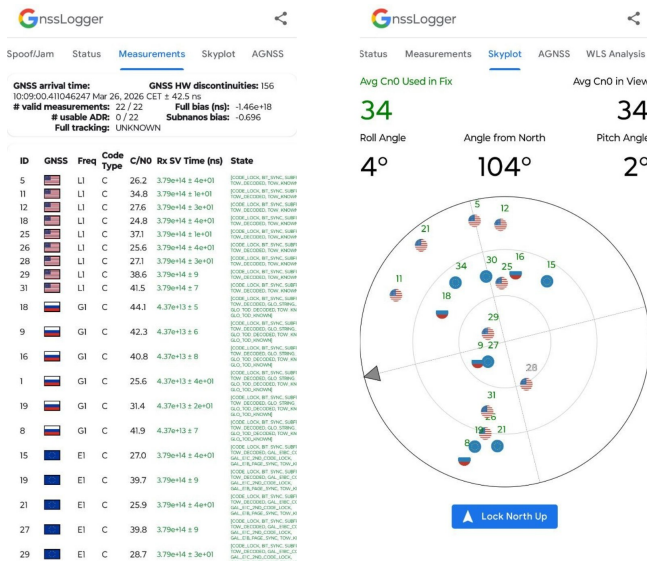


Figure 9: GNSSLogger measurements and skyplot at Comala Association (TO).

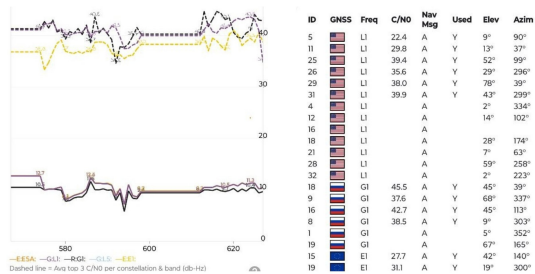


Figure 10: GNSSLogger spoof/jam and status at Comala Association (TO).

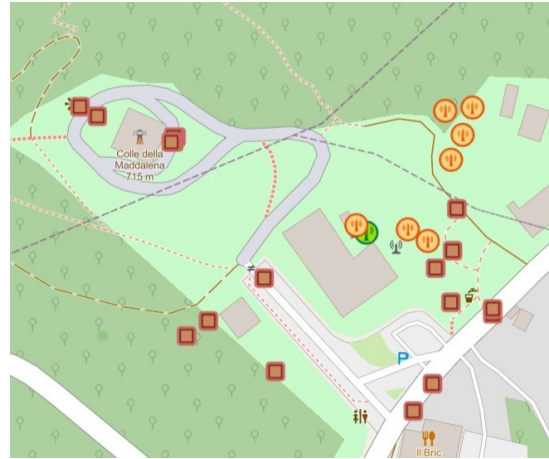


Figure 11: Arpa Piemonte. RF condition at Colle della Maddalena (TO).



Figure 12: DVB-T/FM antennas at Colle della Maddalena (TO). [Credits: Carlo Chanoux]

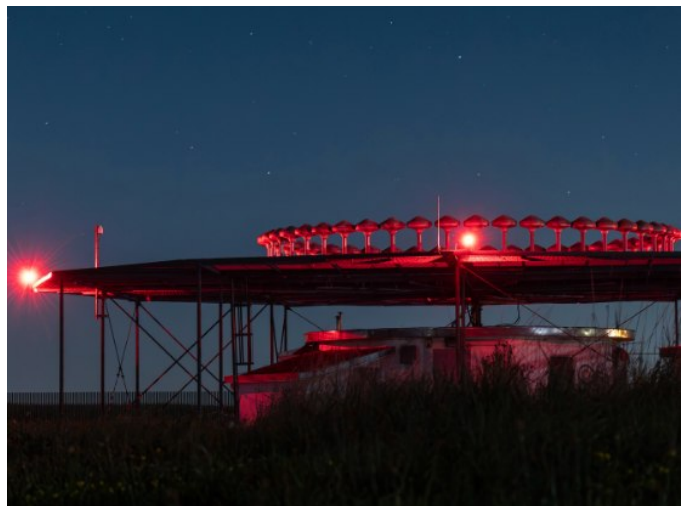


Figure 13: TOP VOR/DME in Porino (TO). [Credits: Instagram @1836.av]